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GATE ECE · CONTROL SYSTEMS · CHAPTER-WISE PYQS

Frequency Response Analysis

GATE ECE Control Systems · Previous Year Questions with Answer Key

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Marks 2	37 questions
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Marks 10	1 question
Answer key	at the end

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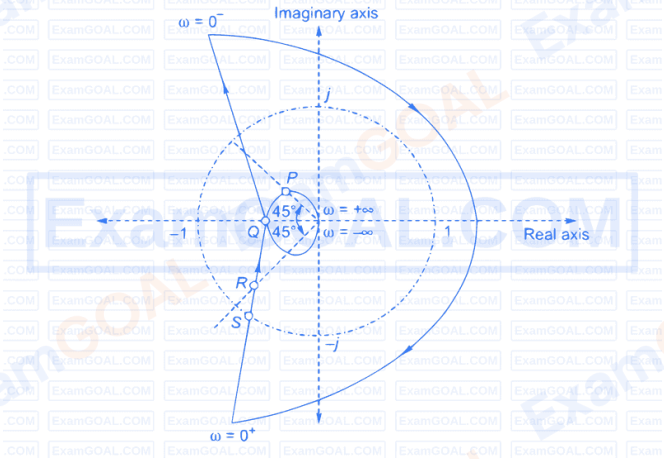
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Marks 1 31 questions

Q1. GATE ECE 2025

The Nyquist plot of a system is given in the figure below. Let $\omega_P, \omega_Q, \omega_R$, and ω_S be the positive frequencies at the points P, Q, R, and S, respectively. Which one of the following statements is TRUE?

Open solution



(A) ω_s is the gain crossover frequency and ω_p is the phase crossover frequency

(B) ω_Q is the gain crossover frequency and ω_R is the phase crossover frequency

(C) ω_Q is the gain crossover frequency and ω_S is the phase crossover frequency

(D) ω_S is the gain crossover frequency and ω_Q is the phase crossover frequency

Q2. GATE ECE 2024

In the context of Bode magnitude plots, 40 dB/decade is the same as _____.

(A) 12 dB/octave (B) 6 dB/octave (C) 20 dB/octave (D) 10 dB/octave

Open solution

Q3. GATE ECE 2023

The open loop transfer function of a unity negative feedback system is $G(s) = \frac{k}{s(1+sT_1)(1+sT_2)}$, where k, T_1 and T_2 are positive constants. The phase cross-over frequency, in rad/s, is

(A) $\frac{1}{\sqrt{T_1 T_2}}$ (B) $\frac{1}{T_1 T_2}$ (C) $\frac{1}{T_1\sqrt{T_2}}$ (D) $\frac{1}{T_2\sqrt{T_1}}$

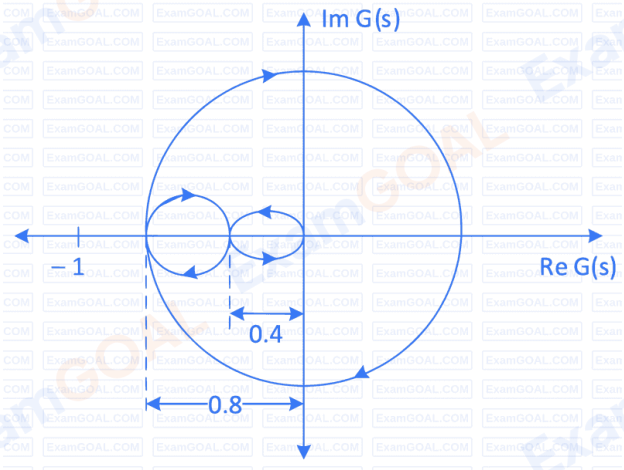
Open solution

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Q4. GATE ECE 2022

Open solution

Consider a closed-loop control system with unity negative feedback and KG(s) in the forward path, where the gain K = 2. The complete Nyquist plot of the transfer function G(s) is shown in the figure. Note that the Nyquist contour has been chosen to have the clockwise sense. Assume G(s) has no poles on the closed right-half of the complex plane. The number of poles of the closed-loop transfer function in the closed right-half of the complex plane is _____.



(A) 0 (B) 1 (C) 2 (D) 3

Q5. GATE ECE 2016 Set 2

Open solution

The number and direction of encirclements around the point −1 + j0 in the complex plane by the Nyquist plot of G(s) = $\frac{1-s}{4+2s}$ is

(A) zero
(B) one, anti-clockwise
(C) one, clockwise
(D) two, clockwise

Q6. GATE ECE 2016 Set 1

Open solution

A closed-loop control system is stable if the Nyquist plot of the corresponding open-loop transfer function

(A) encircles the s-plane point (−1 + j0) in the counterclockwise direction as many times as the number of right-half s-plane poles.
(B) encircles the s-plane point (0 − j1) in the clockwise direction as many times as the number of right-half s-plane poles.
(C) encircles the s-plane point (−1 + j0) in the counterclockwise direction as many times as the number of left-half s-plane poles.
(D) encircles the s-plane point (−1 + j0) in the counterclockwise direction as many times as the number of right-half s-plane zeros.

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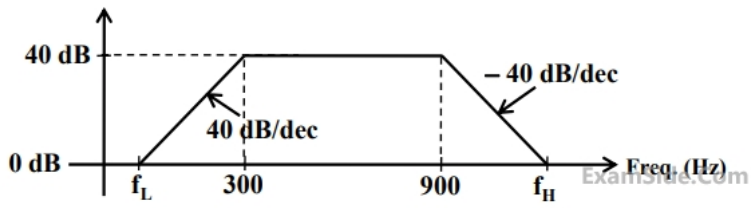
Q7. GATE ECE 2015 Set 3

The phase margin (in degrees) of the system G(s)=$\frac{10}{(s+10)}$ is _____.

Open solution

Q8. GATE ECE 2015 Set 3

Consider the Bode plot shown in the figure. Assume that all the poles and zeroes are real-valued. The value of f_H - f_L (in Hz) is _____.



Open solution

Q9. GATE ECE 2015 Set 1

The polar plot of the transfer function G(s)=$\frac{10(s+1)}{(s+10)}$ for $0 \leq \omega < \infty$ will be in the

(A) first quadrant (B) second quadrant (C) third quadrant (D) fourth quadrant

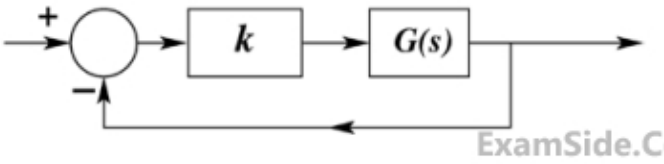
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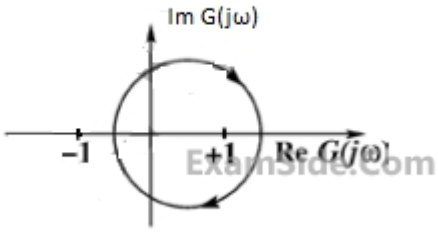
Q10. GATE ECE 2014 Set 1

Consider the feedback system shown in the figure. The Nyquist plot of G(s) is also shown. Which one of the following conclusions is correct?

Open solution



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(A) G(s) is an all-pass filter.

(B) G(s) is a strictly proper transfer function.

(C) G(s) is a stable and minimum-phase transfer function.

(D) The closed-loop system is unstable for sufficiently large and positive k.

Q11. GATE ECE 2014 Set 4

In a Bode magnitude plot, which one of the following slopes would be exhibited at high frequencies by a 4th order all-pole system?

Open solution

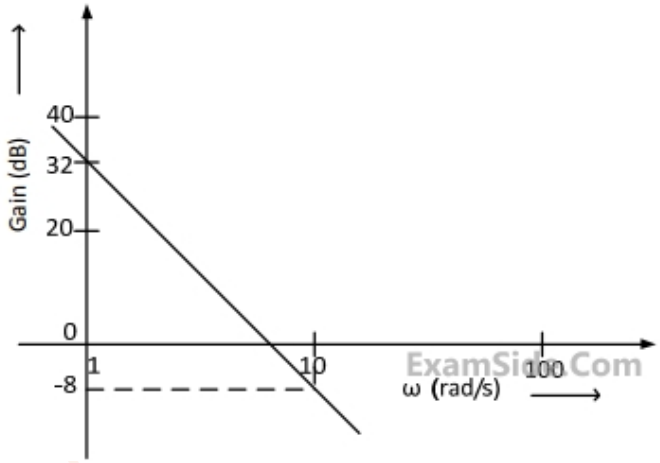
(A) − 80 dB/decade (B) − 40 dB/decade (C) +40 dB/decade (D) +80 dB/decade

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Q12. GATE ECE 2013

Open solution

The Bode plot of a transfer function G (s) is shown in the figure below.



The gain (20 log $|G(s)|$) is 32 dB and -8dB at 1rad/s and 10rad/s respectively. The phase is negative for all ω. Then G(s) is

(A) $\frac{39.8}{s}$

(B) $\frac{39.8}{s^2}$

(C) $\frac{32}{s}$

(D) $\frac{32}{s^2}$

Q13. GATE ECE 2012

Open solution

A system with transfer function g(s) = $\frac{(s^2+9)(s+2)}{(s+1)(s+3)(s+4)}$, is excited by $\sin(\omega t)$. The steady-state output of the system is zero at

(A) $\omega = 1rad/\sec$

(B) $\omega = 2rad/\sec$

(C) $\omega = 3rad/\sec$

(D) $\omega = 4rad/\sec$

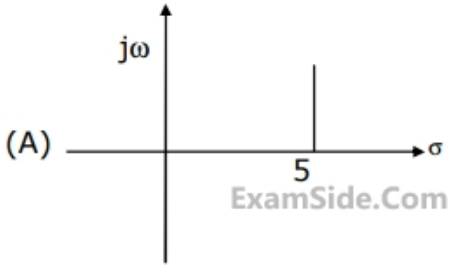
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Q14. GATE ECE 2011

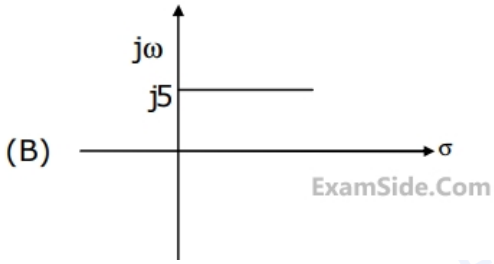
For the transfer function G$(j\omega) = 5 + j\omega$, the corresponding Nyquist plot for positive frequency has the form

Open solution

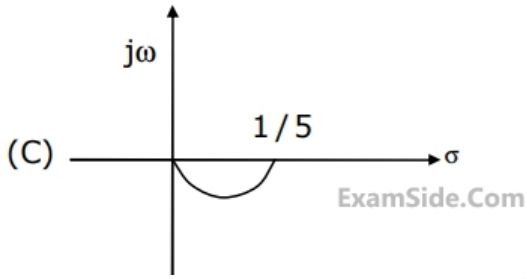
(A)



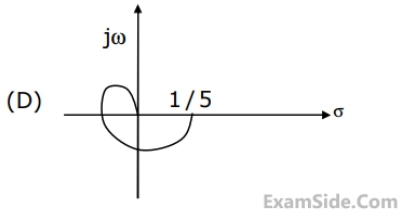
(B)



(C)



(D)



Q15. GATE ECE 2010

A system with the transfer function $\frac{Y(s)}{X(s)} = \frac{s}{s+p}$, has an output y(t)=$\cos\left(2t - \frac{\pi}{3}\right)$, for input signal x(t)=$p\cos\left(2t - \frac{\pi}{2}\right)$. Then the system parameter 'p' is

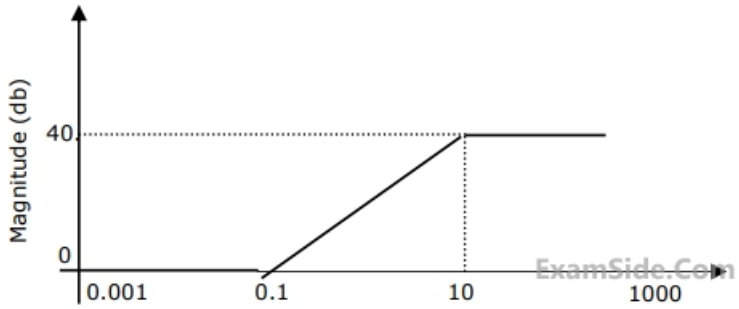
Open solution

(A) $\sqrt{3}$ (B) $\frac{2}{\sqrt{3}}$ (C) 1 (D) $\frac{\sqrt{3}}{2}$

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Q16. GATE ECE 2010

For the asymptotic Bode magnitude plot shown below, the system transfer function can be



(A) $\frac{10s+1}{0.1s+1}$ (B) $\frac{100s+1}{0.1s+1}$ (C) $\frac{100s}{10s+1}$ (D) $\frac{0.1s+1}{10s+1}$

Open solution

Q17. GATE ECE 2007

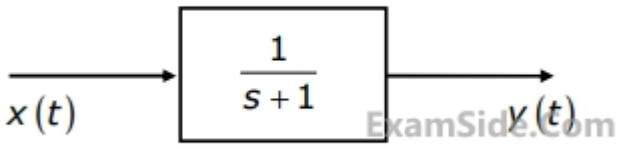
If the closed-loop transfer function of a control system is given as T(s)=$\frac{s-5}{(s+2)(s+3)}$, then it is

(A) an unstable system
(B) an uncontrollable system
(C) a minimum phase system
(D) a non-minimum phase system

Open solution

Q18. GATE ECE 2006

In the system shown below, x(t)=(sin t). In steady-state, the response y(t) will be



(A) $\frac{1}{\sqrt{2}}\sin\left(t-\frac{\pi}{4}\right)$ (B) $\frac{1}{\sqrt{2}}\sin\left(t+\frac{\pi}{4}\right)$
(C) $\frac{1}{\sqrt{2}}e^{-t}\sin t$ (D) $\sin t-\cos t$

Open solution

Q19. GATE ECE 2006

The open-loop transfer function of a unity-gain feedback control system is given by $G(s)=\frac{K}{(s+1)(s+2)}$, the gain margin of the system in dB is given by

(A) 0 (B) 1 (C) 20 (D) ∞

Open solution

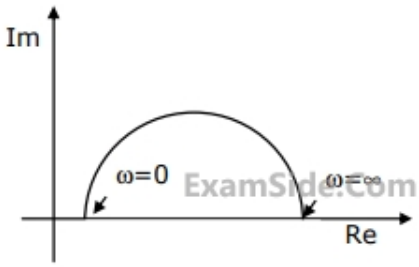
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Q20. GATE ECE 2005

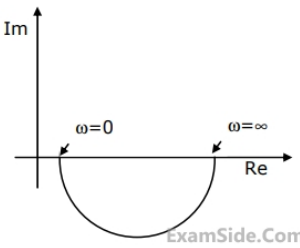
Open solution

Which one of the following polar diagrams corresponds to a lag network?

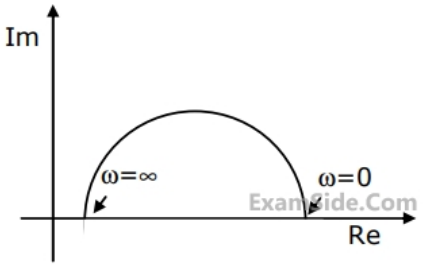
(A)



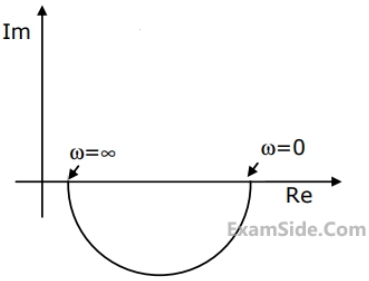
(B)



(C)



(D)

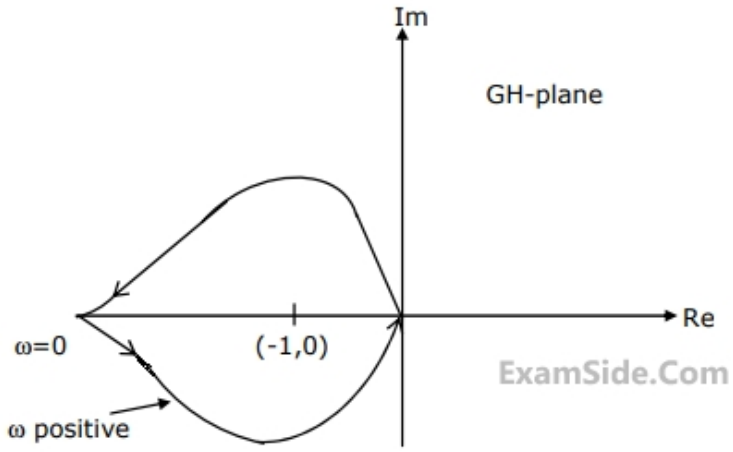


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Q21. GATE ECE 2003

Figure shows the Nyquist plot of the open-loop transfer function G(s)H(s) of a system. If G(s)H(s) has one right hand pole, the closed loop system is

Open solution



(A) always stable

(B) unstable with one closed-loop right hand pole

(C) unstable with two closed-loop right hand poles

(D) unstable with three closed-loop right hand poles

Q22. GATE ECE 2003

The gain margin for the system with open-loop transfer function G(s)H(s)=$\frac{2(1+s)}{s^2}$ is

(A) ∞ (B) 0 (C) 1 (D) $-\infty$

Open solution

Q23. GATE ECE 2002

The phase margin of a system with the open-loop transfer function G(s)H(s)=$\frac{(1-s)}{(1+s)(2+s)}$ is?

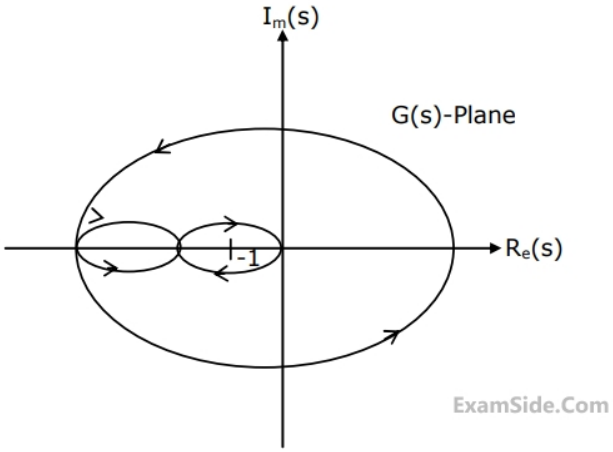
(A) 0° (B) 63.4° (C) 90° (D) ∞

Open solution

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Q24. GATE ECE 2001

The Nyquist plot for the open-loop transfer function G(s) of a unity negative feedback system is shown in figure. if G(s) has no pole in the right half of s-plane, the number of roots of the system characteristic equation in the right half of s-plane is



Open solution

(A) 0 (B) 1 (C) 2 (D) 3

Q25. GATE ECE 1999

The gain margin of a system having the loop transfer function G(s)H(s) = $\frac{\sqrt{2}}{s(s+1)}$ is

(A) 0 (B) 3 (C) 6 (D) ∞

Open solution

Q26. GATE ECE 1999

The phase margin (in degrees) of a system having the loop transfer function is $G(s)H(s) = \frac{2\sqrt{3}}{s(s+1)}$

(A) 45° (B) -30° (C) 60° (D) 30°

Open solution

Q27. GATE ECE 1998

In the Bode-plot of a unity feedback control system, the value of phase of G($j\omega$) at the gain cross over frequency is -125°. The phase margin of the system is

(A) -125° (B) -55° (C) 55° (D) 125°

Open solution

Q28. GATE ECE 1998

The Nyquist plot of a loop transfer function G($j\omega$) H($j\omega$), of a system encloses the (-1, j0) point. The gain margin of the system is

(A) less than zero (B) zero

(C) greater than zero (D) infinity

Open solution

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Q29. GATE ECE 1995

Non - minimum phase transfer function is defined as the transfer function

(A) which has zeroes in the right - half of S-plane.

(B) which has zeroes only in the left - half of S-plane.

(C) which has poles in the right - half of S-plane.

(D) which has poles in the left - half of S-plane.

Open solution

Q30. GATE ECE 1994

The 3-dB bandwidth of a typical second- order system with the transfer function $\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$, is given by

(A) $\omega_n\sqrt{1-2\xi^2}$

(B) $\omega_n\sqrt{(1-\xi^2)+\sqrt{\xi^4-\xi^2+1}}$

(C) $\omega_n\sqrt{(1-2\xi^2)+\sqrt{4\xi^4-4\xi^2+2}}$

(D) $\omega_n\sqrt{(1-2\xi^2)-\sqrt{4\xi^4-4\xi^2+2}}$

Open solution

Q31. GATE ECE 1994

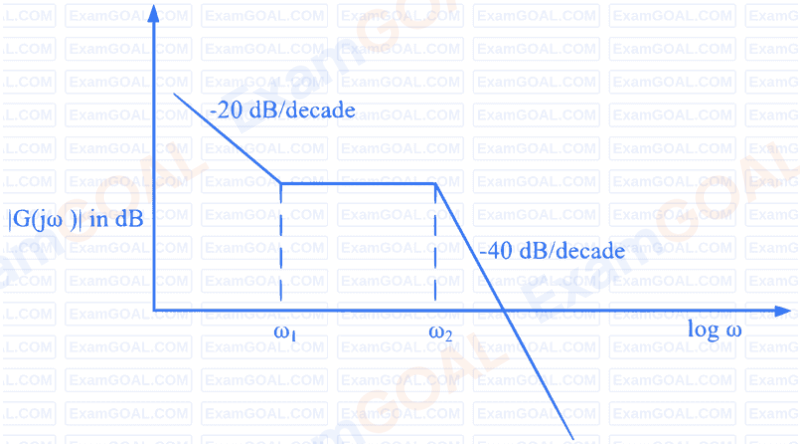
The open loop frequency response of a system at two particular frequencies are given by: 1.2 $\angle -180^\circ$ and 1.0 $\angle -190^\circ$. The closed loop unity feedback control system is then

Open solution

Marks 2 37 questions

Q32. GATE ECE 2023

The asymptotic magnitude Bode plot of a minimum phase system is shown in the figure. The transfer function of the system is $(s) = \frac{k(s+z)^a}{s^b(s+p)^c}$, where k, z, p, z, b and c are positive constants. The value of $(a+b+c)$ is _____ (rounded off to the nearest integer)



Open solution

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Q33. GATE ECE 2020

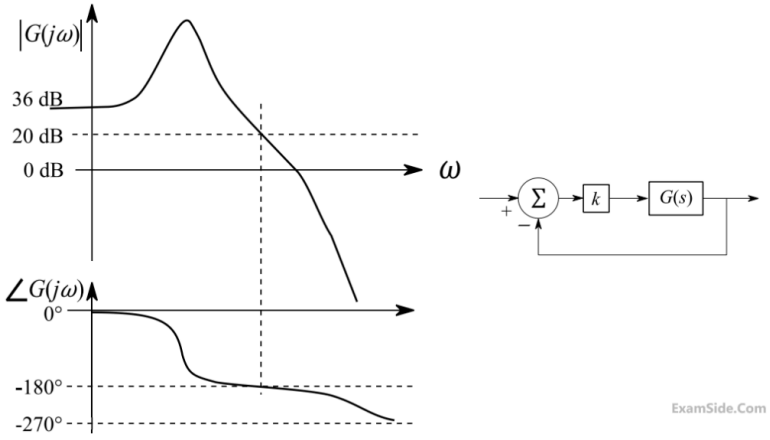
A system with transfer function $G(s) = \frac{1}{(s+1)(s+a)}, a > 0$ is subjected to input $5\cos 3t$. The steady state output of the system is $\frac{1}{\sqrt{10}}\cos(3t - 1.892)$. The value of a is _____.

Open solution

Q34. GATE ECE 2018

The figure below shows the Bode magnitude and phase plots of a stable transfer function

$G(s) = \frac{n_0}{s^3+d_2s^2+d_1s+d_0}$.



Consider the negative unity feedback configuration with gain k in the feedforward path. The closed loop is stable for k < k_0. The maximum value of k_0 is _____.

Open solution

Q35. GATE ECE 2018

For a unity feedback control system with the forward path transfer function

$G(s) = \frac{K}{s(s+2)}$

The peak resonant magnitude M_r of the closed-loop frequency response is 2. The corresponding value of the gain K (correct to two decimal places) is _____.

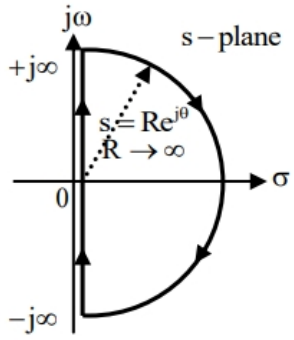
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Q36. GATE ECE 2017 Set 2

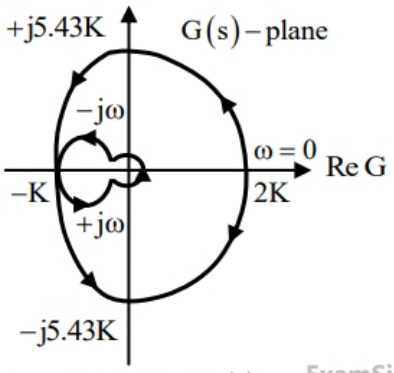
Open solution

A unity feedback control system is characterized by the open loop transfer function $G(s) = \frac{10k(s+2)}{(s^3+3s^2+10)}$ The Nyquist path and the corresponding Nyquist plot of g(s) are shown in the figures below.



Nyquist path for G(s)

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Nyquist Plot of G(s)

ExamSide.Com

If 0 < K < 1, then number of poles of the closed loop transfer function that lie in the right half of the s-plane is

(A) 0 (B) 1 (C) 2 (D) 3

Q37. GATE ECE 2017 Set 1

Open solution

The Nyquist plot of the transfer function $G(s) = \frac{k}{(s^2+2s+2)(s+2)}$ does not encircle the point (-1+j0) for K = 10 but does encircle the point (-1+j0) for K = 100. Then the closed loop system (having unity gain feedback) is

(A) stable for K = 10 and stable for K = 100

(B) stable for K = 10 and unstable for K = 100

(C) unstable for K = 10 and stable for K =100

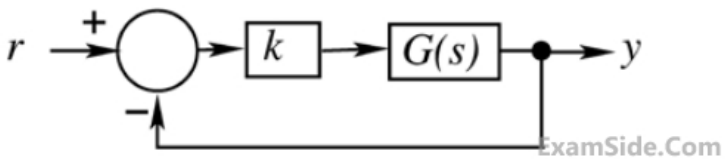
(D) unstable for K = 10 and unstable for K = 100

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Q38. GATE ECE 2016 Set 2

Open solution

In the feedback system shown below $G(s) = \frac{1}{(s+1)(s+2)(s+3)}$



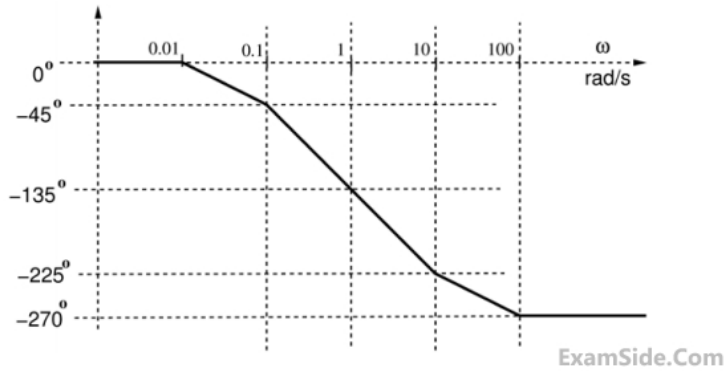
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The positive value of k for which the gain margin of the loop is exactly 0 dB and the phase margin of the loop is exactly zero degree is ____

Q39. GATE ECE 2016 Set 2

Open solution

The asymptotic Bode phase plot of $G(s) = \frac{k}{(s+0.1)(s+10)(s+p_1)}$, with k and p_1 both positive, is shown below.



ExamSide.Com

The value of p_1 is _____

Q40. GATE ECE 2015 Set 2

Open solution

The transfer function of a mass-spring damper system is given by $G(s) = \frac{1}{Ms^2+Bs+K}$ The frequency response data for the system are given in the following table.

ω in rad/s	$\lvert G(j\omega)\rvert$ in dB	arg $(G(j\omega))$ in deg
0.01	-18.5	-0.2
0.1	-18.5	-1.3
0.2	-18.4	-2.6
1	-16	-16.9
2	-11.4	-89.4
3	-21.5	-151
5	-32.8	-167
10	-45.3	-174.5

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The unit step response of the system approaches a steady state value of _____.

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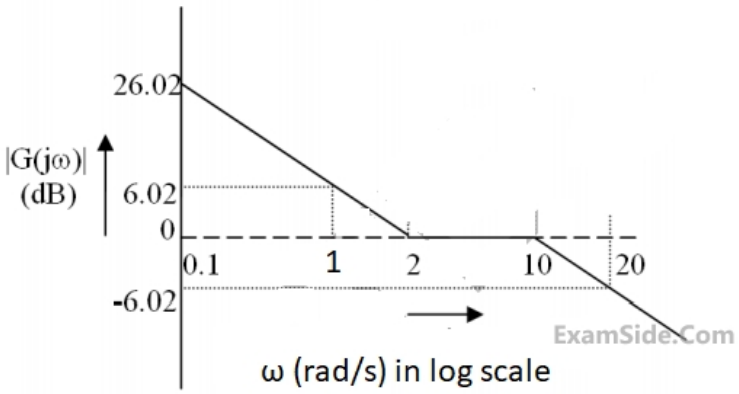
Q41. GATE ECE 2014 Set 1

The phase margin in degrees of G(s)=$\frac{10}{(s+0.1)(s+1)(s+10)}$, using the asymptotic Bode plot is _____

Open solution

Q42. GATE ECE 2014 Set 2

The Bode asymptotic magnitude plot of a minimum phase system is shown in the figure.



Open solution

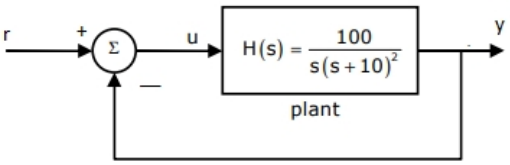
If the system is connected in a unity negative feedback configuration, the steady state error of the closed loop system, to a unit ramp input, is

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Q43. GATE ECE 2011

The input-output transfer function of a plant is $h(s)=\frac{100}{s(s+10)^2}$. The plant is placed in a unity negative feedback configuration as shown in the figure below.

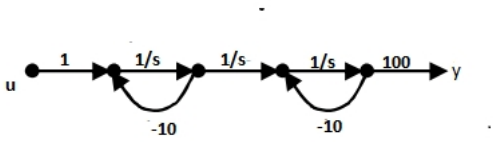
Open solution



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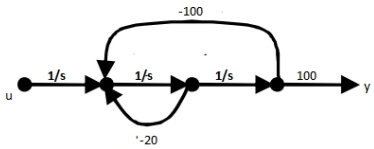
The signal flow graph that DOES NOT model the plant transfer function H(s) is

(A)



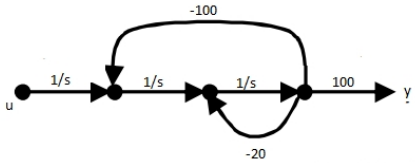
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(B)



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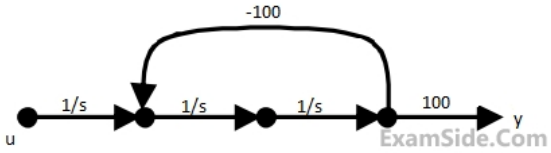
(C)



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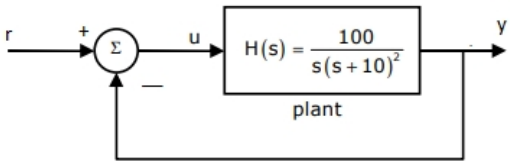
(D)



Q44. GATE ECE 2011

The input-output transfer function of a plant is $h(s)=\frac{100}{s(s+10)^2}$. The plant is placed in a unity negative feedback configuration as shown in the figure below.

Open solution



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The gain margin of the system under closed loop unity negative feedback is

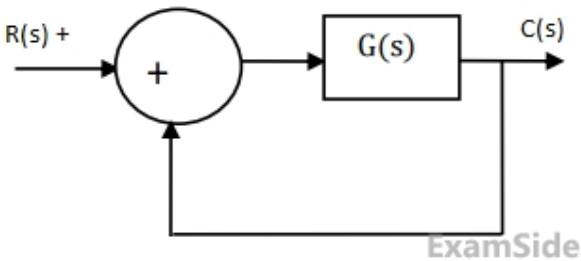
(A) 0 dB (B) 20 dB (C) 26 dB (D) 46 dB

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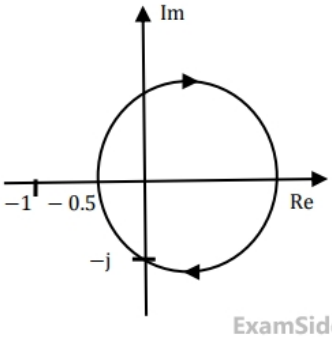
Q45. GATE ECE 2009

The Nyquist plot of a stable transfer function G(s) is shown in the figure. We are interested in the stability of the closed loop system in the feedback configuration shown.

Open solution



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Which of the foloowing statements is true?

(A) G(s) is is an all-pass filter.

(B) G(s) has a zero in the right-half of S-plane.

(C) G(s) is the impedance of a passive network.

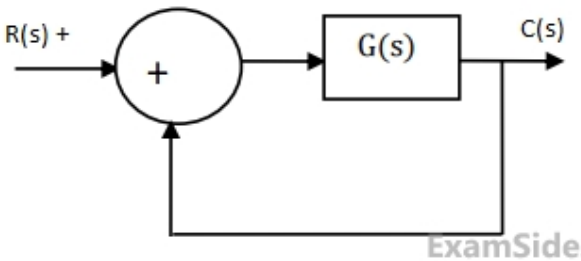
(D) G(s) is marginally stable.

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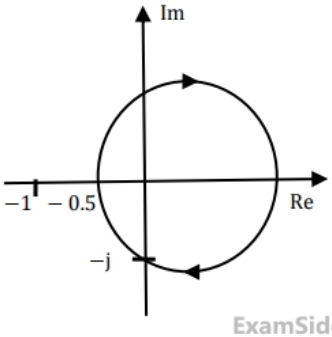
Q46. GATE ECE 2009

The Nyquist plot of a stable transfer function G(s) is shown in the figure. We are interested in the stability of the closed loop system in the feedback configuration shown.

Open solution



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The gain and phase margins of G(s) for closed loop stability are

(A) 6 dB and 180° (B) 3 dB and 180° (C) 6 dB and 90° (D) 3 dB and 90°

Q47. GATE ECE 2008

The magnitude of frequency response of an underdamped second order system is 5 at 0 rad/sec and peaks to $\frac{10}{\sqrt{3}}$ at 5 $\sqrt{2}$ rad/sec. The transfer function of the system is

Open solution

(A) $\frac{500}{s^2+10s+100}$ (B) $\frac{375}{s^2+5s+75}$

(C) $\frac{720}{s^2+12s+144}$ (D) $\frac{1125}{s^2+25s+225}$

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Q48. GATE ECE 2008

The impulse response h(t) of a linear time invariant system is given by h(t) = $e^{-2t}u(t)$, where u(t) denotes the unit step function.
The frequency response H(ω) of the system in terms of angular frequency 'ω' is given by h(ω)

(A) $\frac{1}{1+j2\omega}$
(B) $\frac{\sin\omega}{\omega}$
(C) $\frac{1}{2+j\omega}$
(D) $\frac{j\omega}{2+j\omega}$

Open solution

Q49. GATE ECE 2008

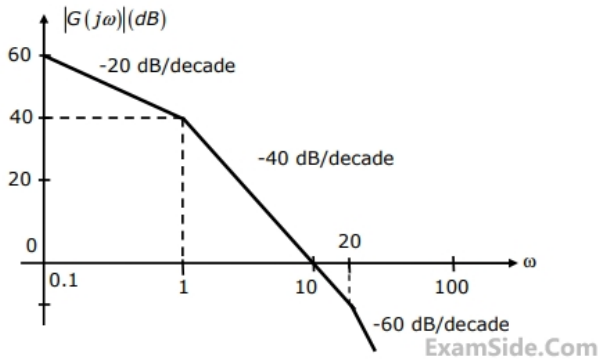
The impulse response h(t) of a linear time invariant system is given by h(t) = $e^{-2t}u(t)$, where u(t) denotes the unit step function.
The output of this system to the sinusoidal input x(t) = 2cos(2t) for all time 't' is

(A) 0
(B) $2^{-0.25}\cos(2t-0.125\pi)$
(C) $2^{-0.5}\cos(2t-0.125\pi)$
(D) $2^{-0.5}\cos(2t-0.25\pi)$

Open solution

Q50. GATE ECE 2007

The asymptotic Bode plot of a transfer function is shown in the figure. the transfer function G(s) corresponding to this bode plot is



(A) $\frac{1}{(s+1)(s+20)}$
(B) $\frac{1}{s(s+1)(s+20)}$
(C) $\frac{100}{s(s+1)(s+20)}$
(D) $\frac{100}{s(s+1)(1+0.05s)}$

Open solution

Q51. GATE ECE 2006

Consider two transfer functions $G_1(s)=\frac{1}{s^2+as+b}$ and $G_2(s)=\frac{s}{s^2+as+b}$. The 3-dB bandwidths of their frequency responses are, respectively

(A) $\sqrt{a^2-4b}$, $\sqrt{a^2+4b}$,
(B) $\sqrt{a^2-4b}$, $\sqrt{a^2-4b}$,
(C) $\sqrt{a^2+4b}$, $\sqrt{a^2-4b}$,
(D) $\sqrt{a^2+4b}$, $\sqrt{a^2+4b}$,

Open solution

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Q52. GATE ECE 2006

The Nyquist plot of G(jω)H(jω) for a closed loop control system, passes through (-1,j0) point in the GH plane. The gain margin of the system in dB is equal to

(A) infinite (B) greater than zero

(C) less than zero (D) zero

Open solution

Q53. GATE ECE 2006

Consider a unity-gain feedback control system whose open-loop transfer function is G(s)=$\frac{as+1}{s^2}$ The value of 'a', so that the system has a phase-margin equal to $\pi/4$ is approximately equal to

(A) 2.40 (B) 1.40 (C) 0.84 (D) 0.74

Open solution

Q54. GATE ECE 2006

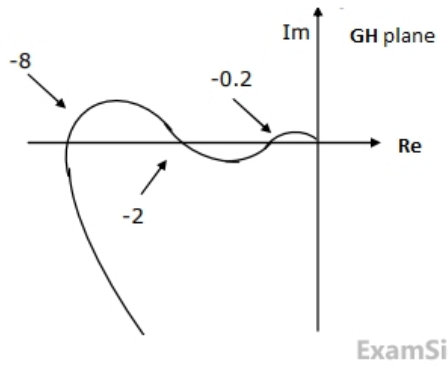
Consider a unity-gain feedback control system whose open-loop transfer function is G(s)=$\frac{as+1}{s^2}$. With the value of "a" set for phase-margin of $\pi/4$, the value of unit-impulse response of the open-loop system at t = 1 second is equal to

(A) 3.40 (B) 2.40 (C) 1.84 (D) 1.74

Open solution

Q55. GATE ECE 2005

The polar diagram of a conditionally stable system for open loop gain K=1 is shown in figure. The open loop transfer function of the system is known to be stable. The closed loop system is stable for



(A) $K < 5$ and $\frac{1}{2} < K < \frac{1}{8}$ (B) $K < \frac{1}{8} and \frac{1}{2} < K < 5$

(C) $K < \frac{1}{8}$ and 5 < K (D) $K > \frac{1}{8}$ and K < 5

Open solution

Q56. GATE ECE 2005

The open loop transfer function of a unity feedback system is given by G(s)=$\frac{3e^{-2s}}{s(s+2)}$. The gain and phase crossover frequencies in rad/sec are, respectively

(A) 0.632 and 1.26 (B) 0.632 and 0.485 (C) 0.485 and 0.632 (D) 1.26 and 0.632

Open solution

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Q57. GATE ECE 2005

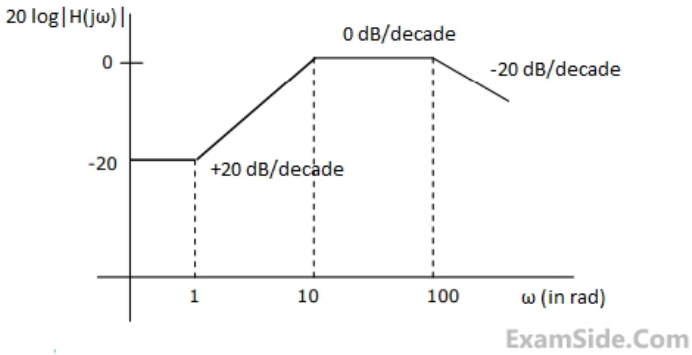
The open loop transfer function of a unity feedback system is given by g(s)=$\frac{3e^{-2s}}{s(s+2)}$. Based on the above results, the gain and phase margins of the system will be

(A) -7.09 dB and 87.5^0
(B) 7.09 dB and 87.5^0
(C) 7.09 dB and -87.5^0
(D) -7.09 dB and -87.5^0

Open solution

Q58. GATE ECE 2004

Consider the Bode magnitude plot shown in figure. The transfer function H(s) is



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(A) $\frac{(s+10)}{(s+1)(s+100)}$
(B) $\frac{10(s+1)}{(s+10)(s+100)}$
(C) $\frac{10^2(s+1)}{(s+10)(s+100)}$
(D) $\frac{10^3(s+100)}{(s+1)(s+10)}$

Open solution

Q59. GATE ECE 2004

A system has poles at 0.01 Hz, 1Hz and 80 Hz; zeroes at 5hz, 100 Hz and 200 Hz. The approximate phase of the system response at 20 Hz is

(A) -90^0
(B) 0^0
(C) 90^0
(D) -180^0

Open solution

Q60. GATE ECE 2003

The gain margin and the phase margin of a feedback system with G(s)H(s)=$\frac{s}{(s+100)^3}$ are

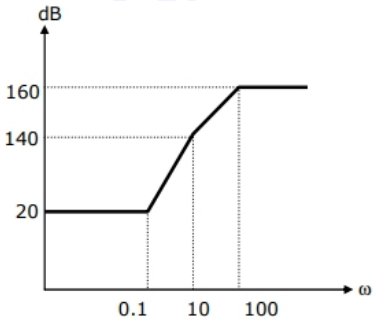
(A) 0 dB, 0^0
(B) ∞, ∞
(C) $\infty, 0^0$
(D) 88.5 dB, ∞

Open solution

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Q61. GATE ECE 2003

The approximate Bode magnitude plot of a minimum-phase system is shown in figure. The transfer function of the system is



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Open solution

(A) $10^8 \frac{(s+0.1)^3}{(s+10)^2(s+100)}$

(B) $10^7 \frac{(s+0.1)^3}{(s+10)(s+100)}$

(C) $10^8 \frac{(s+0.1)^2}{(s+10)^2(s+100)}$

(D) $10^9 \frac{(s+0.1)^3}{(s+10)(s+100)^2}$

Q62. GATE ECE 2002

The system with the open loop transfer function G(s)H(s)= $\frac{1}{s(s^2+s+1)}$, has a gain margin of

(A) -6 dB (B) 0 dB (C) 3.5 dB (D) 6 dB

Open solution

Q63. GATE ECE 2001

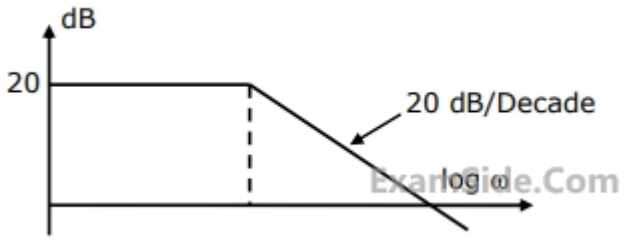
The open-loop DC gain of a unity negative feedback system with closed-loop transfer function $\frac{s+4}{s^2+7s+13}$ is

(A) $\frac{4}{13}$ (B) $\frac{4}{9}$ (C) 4 (D) 13

Open solution

Q64. GATE ECE 1992

Bode plot of a stable system is shown in fig. The transfer function of the system is



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Q65. GATE ECE 1991

The open-loop transfer function of a feedback control system is $G(s)=\frac{1}{(s+1)^3}$

The gain margin of the system is

(A) 2 (B) 4 (C) 8 (D) 16

Open solution

Q66. GATE ECE 1989

From the Nicholas chart, one can determine the following quantities pertaining to a closed loop system:

(A) Magnitude and phase (B) band width

(C) only magnitude (D) only phase

Open solution

Q67. GATE ECE 1987

A system has fourteen poles and two zeroes. Its high frequency asymptote, in its magnitude plot, has having a slope of

(A) -40 dB / decade (B) -240 dB / decade

(C) -280 dB / decade (D) -320 dB / decade.

Open solution

Q68. GATE ECE 1987

The polar plot of $G(s)=\frac{10}{s(s+1)^2}$, intercepts real axis at $\omega=\omega_0$ Then, the real part and ω_0 are respectively given by

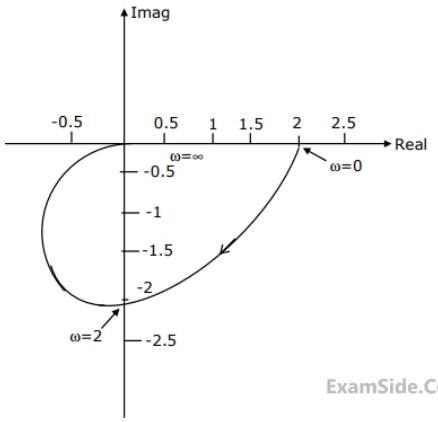
(A) -25, 1 (B) -5, 0.5 (C) -5, 1 (D) -5, 2

Open solution

Marks 5 4 questions

Q69. GATE ECE 2002

The Nyquist plot of an all-pole second order open-loop system is shown in Figure. Obtain the transfer function of the system.



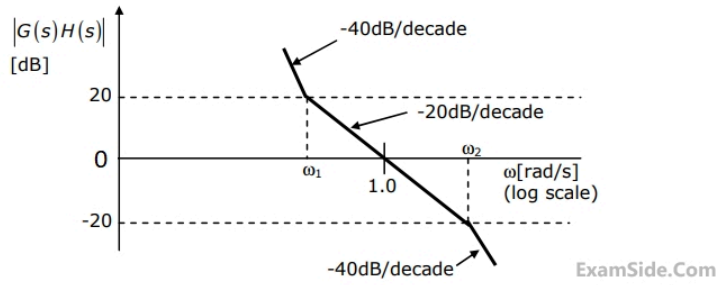
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Q70. GATE ECE 1999

The asymptotic bode plot of the minimum phase open-loop transfer function G(s)H(s) is as shown in the figure. Obtain the transfer function G(s)H(s)



Open solution

Q71. GATE ECE 1999

Consider a feedback system with the open loop transfer function given by $G(s)H(s) = \frac{K}{s(2s+1)}$. Examine the stability of the closed-loop system using Nyquist stability.

Open solution

Q72. GATE ECE 1998

The loop transfer function of a single loop control system is given by $G(s)H(s) = \frac{100}{s(1+0.01s)}e^{-ST}$ Using Nyquist criterion, find the condition for the closed loop system to be stable.

Open solution

Marks 8 2 questions

Q73. GATE ECE 1992

A unity feedback system has open-loop transfer function $G(s) = \frac{1}{s(2s+1)(s+1)}$ Sketch Nyquist plot for the system and there from obtain the gain margin and the phase margin.

Open solution

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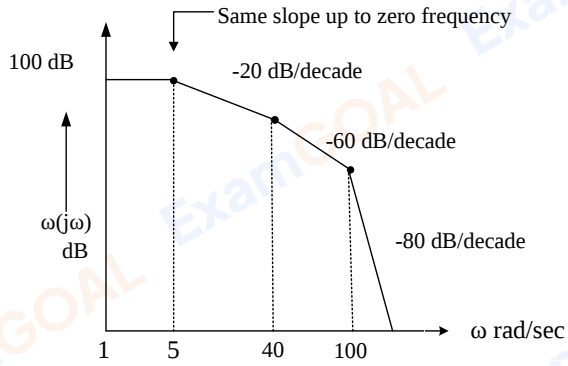
Q74. GATE ECE 1987

The magnitude plot of the open loop transfer function G(s), of a certain system is given in fig,

(a) Determine G(s), if it is known that the system is of minimum phase type.

(b) Estimate the phase at each of the corner frequencies.

Open solution



Marks 10 1 questions

Q75. GATE ECE 1989

Nyquist plot consider a feed back system where the OLTF is: $G(s) = \frac{1}{s(2s+1)(s+1)}$. Determine the asymptote which the nyquist plot approaches as $\omega \to 0$. Find also the range of 'k' in terms of the cross over frequency ω_{PC} for stability.

Open solution

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Answer Key

Marks 1

1. **D**	2. **A**	3. **A**	4. **C**	5. **A**
6. **A**	7. **180**	8. **8970**	9. **A**	10. **D**
11. **A**	12. **B**	13. **C**	14. **A**	15. **B**
16. **A**	17. **D**	18. **A**	19. **D**	20. **D**
21. **A**	22. **A**	23. **D**	24. **A**	25. **D**
26. **D**	27. **C**	28. **A**	29. **A**	30. **C**
31. **Unstable**				

Marks 2

32. **4**	33. **4**	34. **0.1 to 0.1**	35. **14 to 17**	36. **C**
37. **B**	38. **60**	39. **1**	40. **0.11 to 0.12**	41. **45**
42. **0.5**	43. **D**	44. **C**	45. **B**	46. **C**
47. **A**	48. **C**	49. **D**	50. **D**	51. **C**
52. **D**	53. **C**	54. **C**	55. **B**	56. **D**
57. **D**	58. **C**	59. **A**	60. **B**	61. **A**
62. **B**	63. **B**	64. **T.F=** $\frac{10}{(1+s)}$	65. **C**	66. **A**
67. **B**	68. **C**			

Marks 5

69. $G(s) = \frac{8}{(s^2+2s+4)}$	70. $G(s)H(s) = \frac{10(s+0.1)}{s^2(s+10)}$	71. **Solve it.**	72. **When T > 0.011 see system is stable.**

Marks 8

73. **Gain Margin =** 1.49° **Phase Margin =**11.57°	74. **(a)** **G(s)=** $\frac{8\times10^{10}}{(s+5)(s+40)^2(5+100)}$ **(b) Phase at ω = 5 rad/sec = -62.11° Phase at ω =40 rad/sec = -194.67° Phase at ω = 100 rad/sec = -268.54°**

Marks 10

75. **Solve it.**

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